

Fundamental Geometry (Part II) Homework

Name
Date
Course

① Classify the angles with the given measures.

a) 100° b) 75° c) 170°

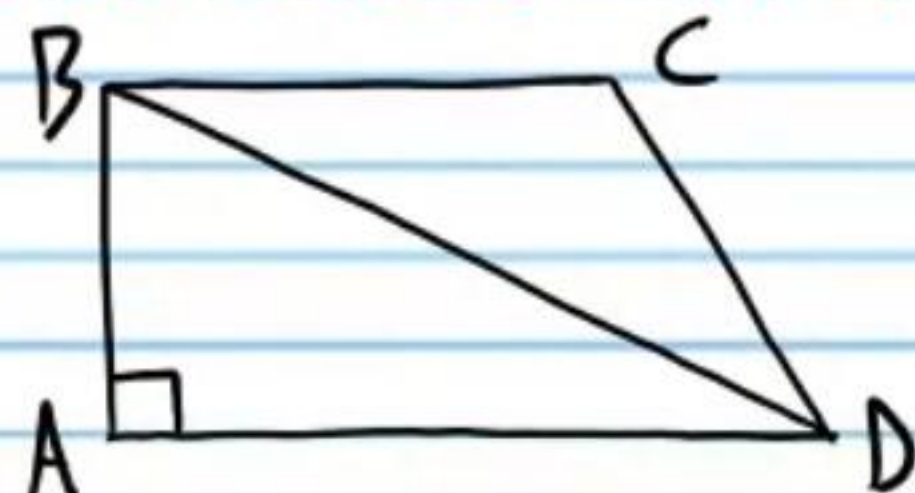
d) 90° e) 180° f) 211°

- a) Obtuse
- b) Acute
- c) obtuse
- d) Right
- e) Straight
- f) Reflex

② Classify each angle below:

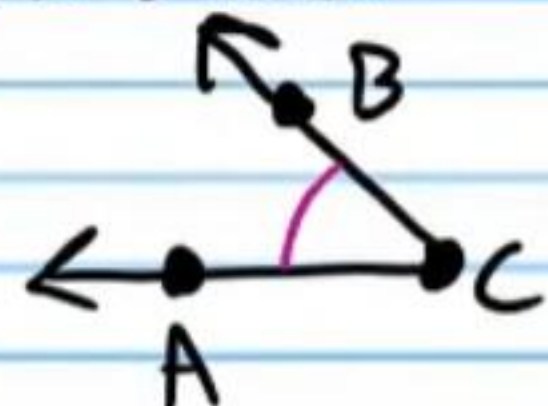
a) $\angle ABD$ b) $\angle BAD$, c) $\angle BCD$, d) $\angle BDA$.

Assume that none of these are reflex angles.



- a) Acute
- b) Right
- c) obtuse
- d) Acute

③ Write three names for the angle below and identify its vertex and sides.

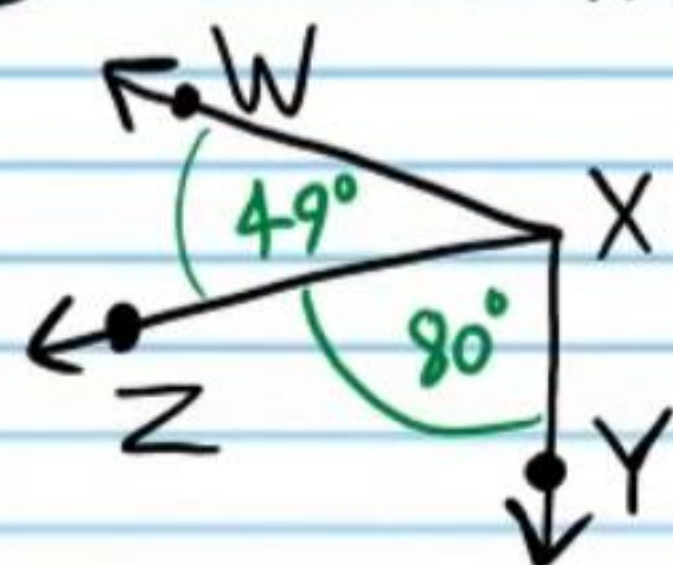


Names
 $\angle ACB$, $\angle BCA$, $\angle C$

Vertex
C

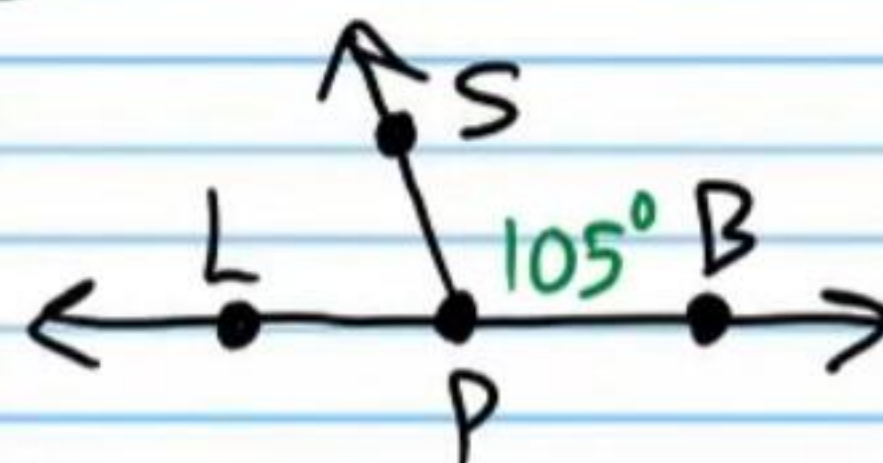
Sides
 $\overrightarrow{CA}$ and $\overrightarrow{CB}$

④ Find $m\angle WXY$



129°

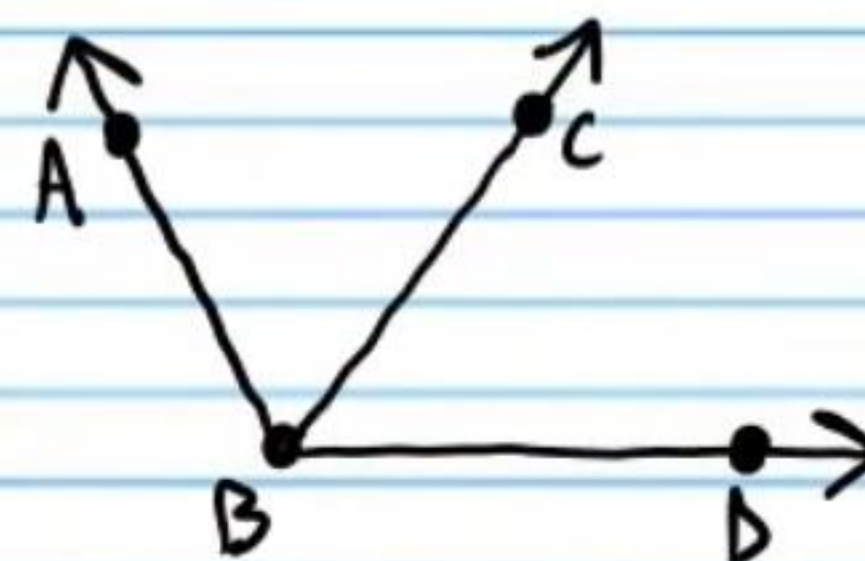
⑤ Find $m\angle LPS$.



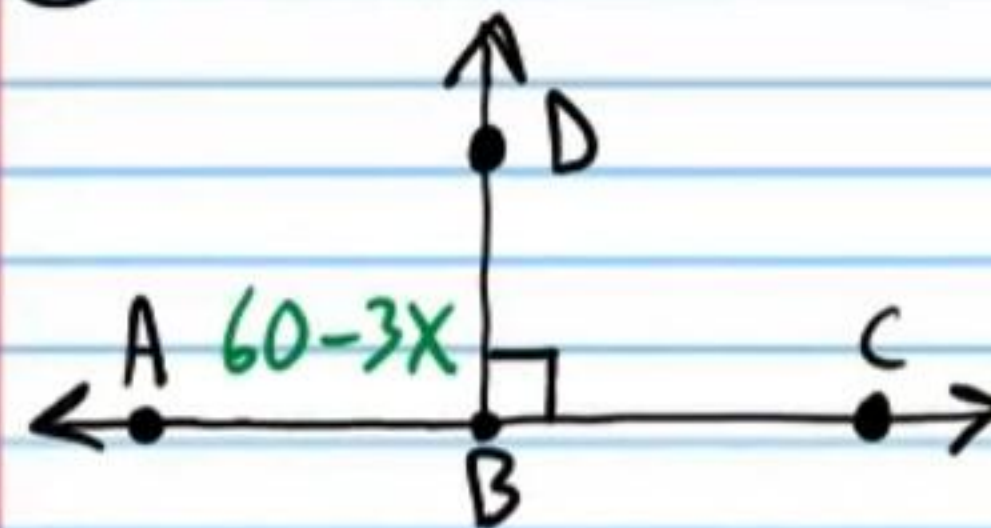
75°

⑥ If $m\angle ABC = 69^\circ$, and $m\angle ABD = 112^\circ$, Find $m\angle CBD$.

43°



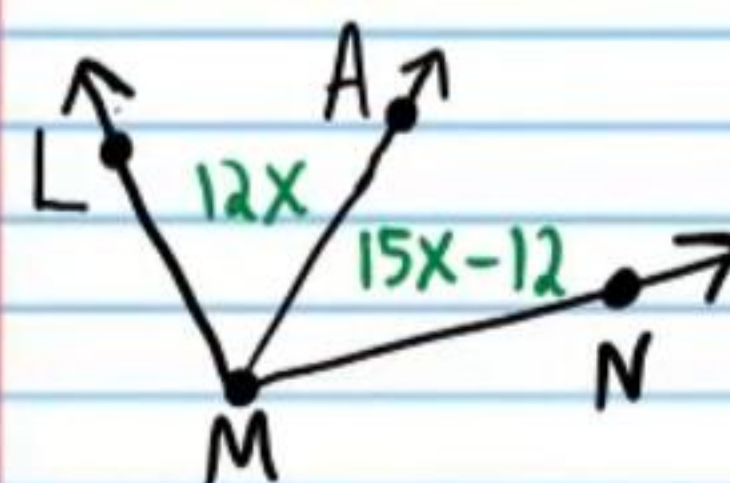
⑦ Find X.



-10

⑧ If $\overrightarrow{MA}$ bisects $\angle LMN$, Find $m\angle AMN$.

48°



9) Classify $\angle ABC$ with $A(-3,2)$, $B(0,8)$, and $C(1,-2)$.
Assume that it is not a reflex angle.

Acute

10) If two angles are complementary, and one of them is 50° , what is the measure of the other angle?

40°

11) If two angles are supplementary, and one of them is 90° , what is the measure of the other angle?

90°

12) If the measure of an angle is twice the measure of its complement, what is the measure of the angle?

60°

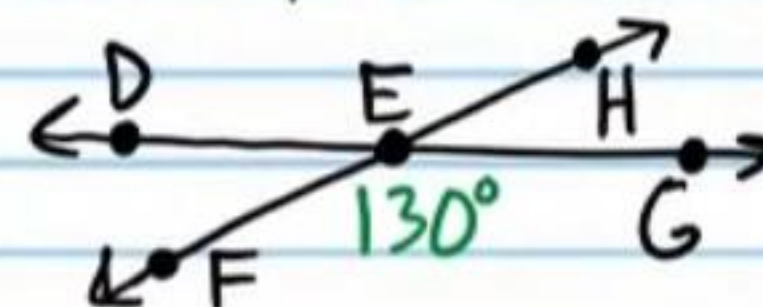
13) If the measure of an angle is 65° less than six times its supplement, what is the measure of the angle?

145°

14) If $\angle A$ and $\angle B$ are supplementary, and $m\angle A = x - 6$ and $m\angle B = 4x + 8$, find x .

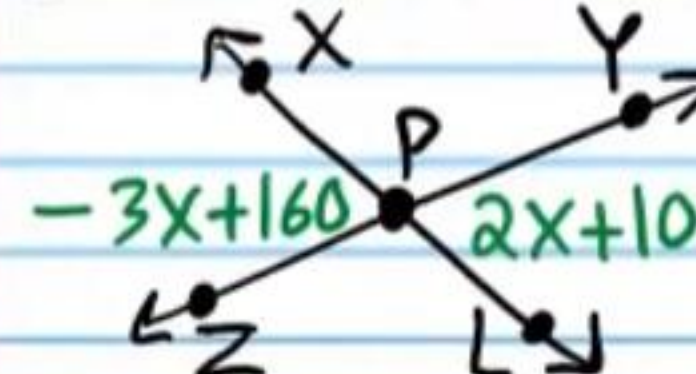
$x = 35.6^\circ$

15) Find $m\angle DEF$, $m\angle HEG$, and $m\angle DEH$.



$m\angle DEF = 50^\circ$
 $m\angle HEG = 50^\circ$
 $m\angle DEH = 130^\circ$

16) Find $m\angle YPL$.



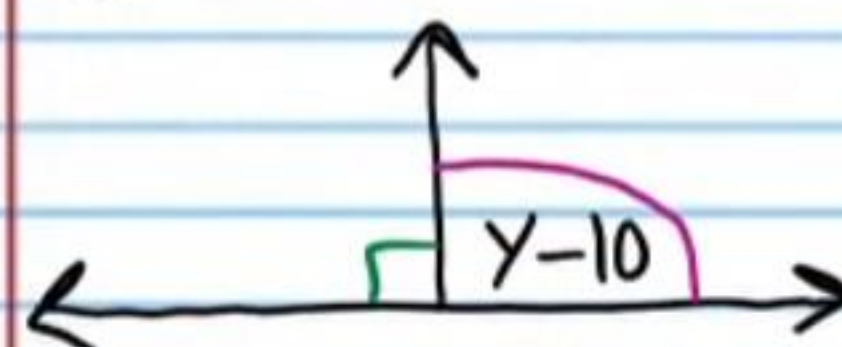
70°

17) a) Identify all pairs of angles below that are adjacent.
b) Identify all pairs of angles below that are nonadjacent.
c) Identify all pairs of angles that are vertical.

a) None
b) $\angle ABC$ & $\angle DEF$.
They do not share a side or a vertex.
c) None

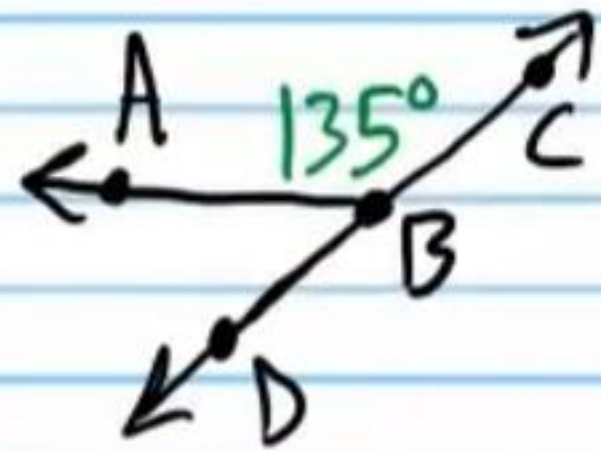


18) Find y .



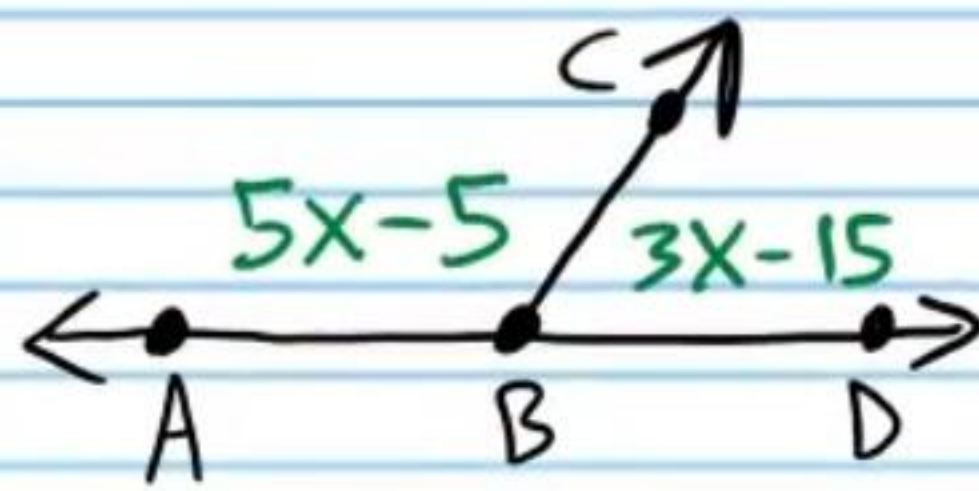
100°

① Find $m\angle ABD$.



45°

② Find $m\angle ABC$.



120°